Tensor Networ methods workshop I
julia langugage basics,
Exact Diagonalisation via XDiag.jl
Transverse Field Ising model and XXZ spin-1/2 model

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Github with codes: `https://github.com/mkpanfil/TN_UW`

Source about `XDiag.jl`:`https://awietek.github.io/xdiag/user_guide/`

Exemplary IDE: `https://vscodium.com/`

Basic setup: Julia language, precompiling, multithreading

Please refer to the file
`https://github.com/mkpanfil/TN_UW/blob/main/julia_MKL_instructions.txt`

XDiag.jl working example

Check if you installed julia and XDiag.jl properly by running
`https://github.com/mkpanfil/TN_UW/blob/main/XDiag_free_fermions_example.jl`.

1 Transverse Field Ising model

The phase diagram of the TFI model looks like this:

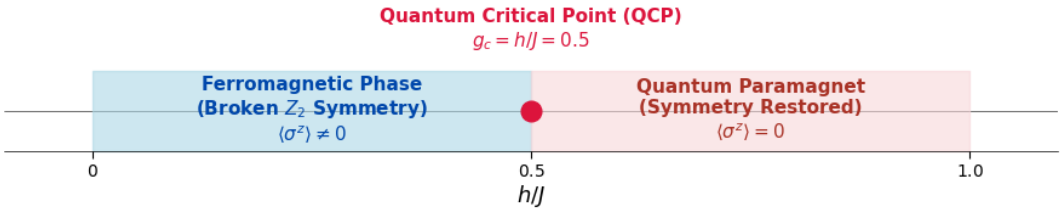


Figure 1: Transverse Field Ising phase diagram

The Hamiltonian of Transverse field Ising model is given by:

$$H_{\mathrm{TFI}} = -J \sum_{i=1}^{L-1} S_i^z S_{i+1}^z - h \sum_{i=1}^{L} S_i^x, \tag{1}$$

where J gives us strength of the interaction in the (z-direction) while h is the applied transverse field (x-direction) and S_i^x, S_i^z are spin-1/2 operators acting on site i, where we have sites going from site 1 up to L.

Spin-1/2 operators can be represented as 2x2 matrices:

$$S_x = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad S_y = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad S_z = \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Which are just Pauli matrices multiplied by $\hbar/2$:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

We take $\hbar \to 1$.

$$H_{\text{OBC}} = -J(S_z \otimes S_z) - h(S_x \otimes I + I \otimes S_x)$$

For a smallest system with open boundary conditions (OBC) the number of sites $L = 2$

$$H_{\text{OBC}} = \begin{pmatrix} -J & -h & -h & 0 \\ -h & J & 0 & -h \\ -h & 0 & J & -h \\ 0 & -h & -h & -J \end{pmatrix}$$

Here the matrix is represented using "computational basis" which is z-resolved. $\{\uparrow\uparrow, \uparrow\downarrow, \downarrow\uparrow, \downarrow\downarrow, \}$.

You could try to diagonalize this matrix, but in overall it grows as 2^L, (where 2 is the size of the spin-1/2 operator) gets out of hand pretty fast. We need to use numerical methods for any serious system.

Methods for Sparse (Block-Diagonal) vs Dense matrices

The most general methods for finding eigenvalues are made for dense matrices (majority or all elements of matrix are nonzero).

$$H_{\text{TFI}}^{\text{OBC}}(L = 3) = \begin{pmatrix} -2J & -h & -h & 0 & -h & 0 & 0 & 0 \\ -h & 0 & 0 & -h & 0 & -h & 0 & 0 \\ -h & 0 & 2J & -h & 0 & 0 & -h & 0 \\ 0 & -h & -h & 0 & 0 & 0 & 0 & -h \\ -h & 0 & 0 & 0 & 0 & -h & -h & 0 \\ 0 & -h & 0 & 0 & -h & 2J & 0 & -h \\ 0 & 0 & -h & 0 & -h & 0 & 0 & -h \\ 0 & 0 & 0 & -h & 0 & -h & -h & -2J \end{pmatrix}$$

The matrix of TFI is actually block-diagonal, so its not exactly a dense matrix (easier to solve numerically).

1.1 Exercise: Find eigenvalues and eigenvectors of TFI matrix via XDiag.jl

Write the code to diagonalize the Ising Hamiltonian with $J = 1, h = 0$ to obtain ground state energy as well as first excited state energy, compute the gap. The functions to diagonalize the matrices are given on the XDiag.jl page: `https://awietek.github.io/xdiag/user_guide/`

The Hamiltonian **without the field** can be defined as:

```
block = Spinhalf(L) #We define the type and number of sites L here
ops = OpSum()

for i in 1:(L - 1)
    ops += -J * Op("SzSz", [i, i+1])
end
```

1.1.1 Save and Plot

Save the data and do a plot: Now do the same for $0 \leq h \leq 1$, plot the gap onto a plot, save plot and plot data to .csv file. Saving looks like this (you can also use dataframes):

```
energy_io = open("energy_data_L$(L).csv", "w")
println(energy_io, "h,E0,E1,Gap")
for (h, E0, E1, gap, c_val) in zip(h_range, E0_vals, E1_vals, gaps)
    println(energy_io, "$h,$E0,$E1,$gap")
end
close(energy_io)
```

While plotting looks like this:

```
plt_gap = plot(h_range, gaps,marker=:circle)
savefig(plt_gap, "TFIM_L$(L)_energy_gap.png")
```

1.1.2 Load and Plot

Load data from .csv and plot it (now for plotting you can also use python). Still in julia you can also just use:

```
energy_df = CSV.read("energy_data_L$(L).csv", DataFrame) #loads csv to dataframe
energy_df.h #this gives you list of fields
energy_df.Gap #this gives you list of gaps
```

1.1.3 Finite size scaling: fitting functions

Select some particular value of a field h, then for this field do energy E_{gap} computations for several system sizes, lets say $L = 4, 6, 8, 10, 12$, plot the E_{gap}/L against the $1/L$. To fit some simple polynomial you can use something like this:

```
using Polynomials
p_fit = fit(x, y, 2)
x_smooth = range(-3, 3, length=200)
y_fit = p_fit.(x_smooth)
```

2 XXZ model

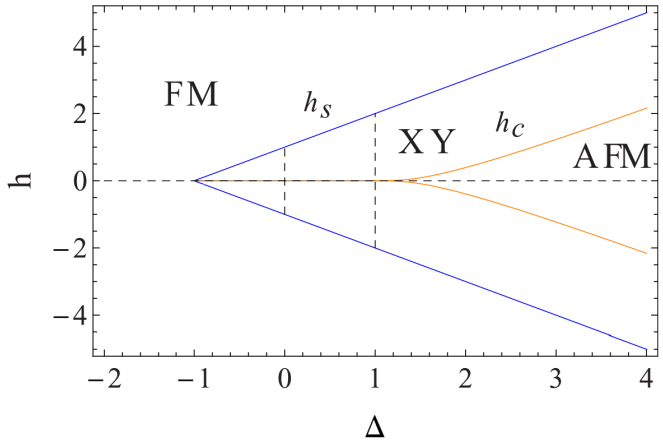


Figure 2: Parrallel field XXZ spin-1/2 model phase diagram
https://journals.aps.org/prb/pdf/10.1103/PhysRevB.93.054417

The Hamiltonian of the XXZ spin-1/2 model is given by:

$$H_{\text{XXZ}} = J(\sum_{i=1}^{L-1} S_i^x S_{i+1}^x + S_i^y S_{i+1}^y + \Delta S_i^z S_{i+1}^z) - h\sum_{i=1}^{L} S_i^x, \tag{2}$$

or alternatively:

$$H_{\text{XXZ}} = J(\sum_{i=1}^{L-1} S_i^+ S_{i+1}^- + S_i^- S_{i+1}^+ + \Delta S_i^z S_{i+1}^z) - h\sum_{i=1}^{L} S_i^x, \tag{3}$$

In XDiag.jl it is implemented as:

```
for i in 1:L-1
    ops += J *Op("Exchange", [i, i+1])
    ops += J *Delta*Op("SzSz", [i, i+1])
end
```

2.1 Exercise: 2D Phase Diagram (can be python also)

Start at $h = 0$ and find phase transitions by varying Δ. Do the same plots as in Transverse Field Ising case.

Then try to do the full phase diagram as seen in Figure 2.

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